AIDEN J. AERNI

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SUMMARY

Mechanical Engineering student focused on design and manufacturing, with hands-on experience developing functional mechanical systems. Seeking a design-focused role within a collaborative team, and known for being reliable, easy to work with, and solutions-driven.

EDUCATION

Bachelor of Science in Mechanical Engineering, *Oregon State University [June 2026]*

- CAD/CAM Design
- Failure Mode Analysis
- Fluid Dynamics
- Heat Transfer
- Mechanics of Materials
- Manufacturing Processes

EXPERIENCE & PROJECTS

4-Axis CNC Product & Business Proposal, *CAD/CAM (MFGE 438)*

- Designed a threaded gas cap in Siemens NX, incorporating tolerances, material selection, and manufacturability considerations.
- Generated CAM toolpaths for a 4-axis lathe/mill and developed a machining plan. Evaluated production feasibility through cost analysis.

Engineering Expo 3kg Combat Robot, *MIME Capstone Design I, II (MIME 497,498)*

- Designed and built a 3 kg SumoBot in SolidWorks, developing a high-torque drivetrain and integrating motor driver circuitry and power systems.
- Engineered a deployable ramp mechanism for offensive engagement and iterated on the design to improve traction, stability, and overall performance.

Servo-Controlled Greenhouse Ventilation (PID System), *Intro to Instrumentation (ME 351)*

- Designed and implemented a servo-controlled system using a PID controller to regulate output based on sensor feedback.
- Integrated sensors and actuators using a microcontroller and tuned control parameters to improve system stability, response time, and accuracy.

WORK EXPERIENCE

Wedding and Events Supervisor, *Black Butte Ranch [June 2021 – Present]*

- Led and coordinated a team of 8-10 staff to execute events of up to 175+ guests, managing task delegation and timelines to ensure on-time, high-quality delivery in fast-paced environments.
- Resolved real-time operational issues during events, adapting to changing conditions and minimizing disruptions to maintain performance and meet strict deadlines.

SKILLS

- **Design and Engineering Thinking:** DFM, tolerance analysis, iterative problem solving.
- **Manufacturing and Prototyping:** CNC machining, CAM toolpaths, rapid prototyping.
- **CAD and Technical Design:** Siemens NX, SolidWorks, assemblies, drawings.
- **Controls and System Integration:** PID control, sensors, microcontrollers.
- **Problem Solving and Debugging**: Troubleshooting mechanical and electromechanical systems.
- **Collaboration and Communication:** Team-based design, reliable and easy to work with.